NMOS CHARACTERIZATION

Cadence Virtuoso / Spectre simulation

STUDENT

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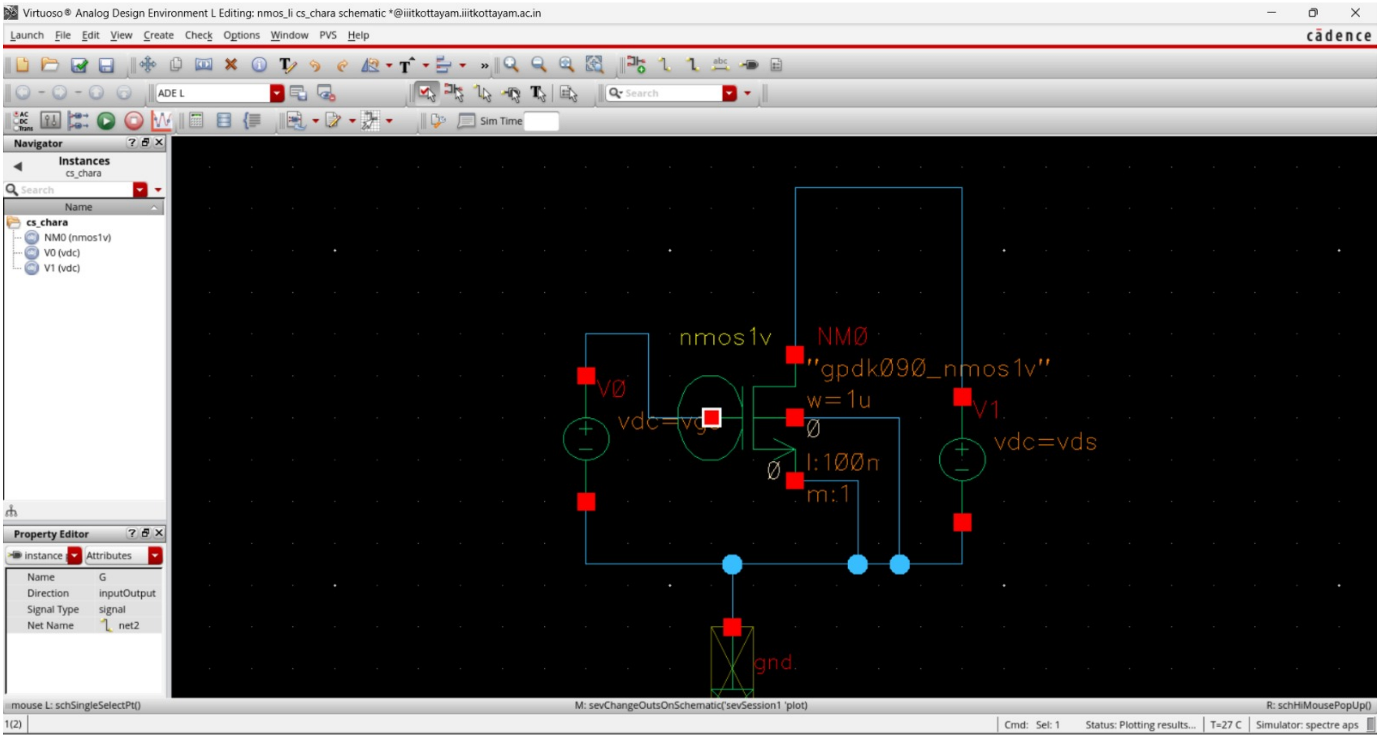
2025BEC0060

REPOSITORY

Schematic

NMOS test circuit



NMOS characterization test circuit implemented in Cadence Virtuoso.

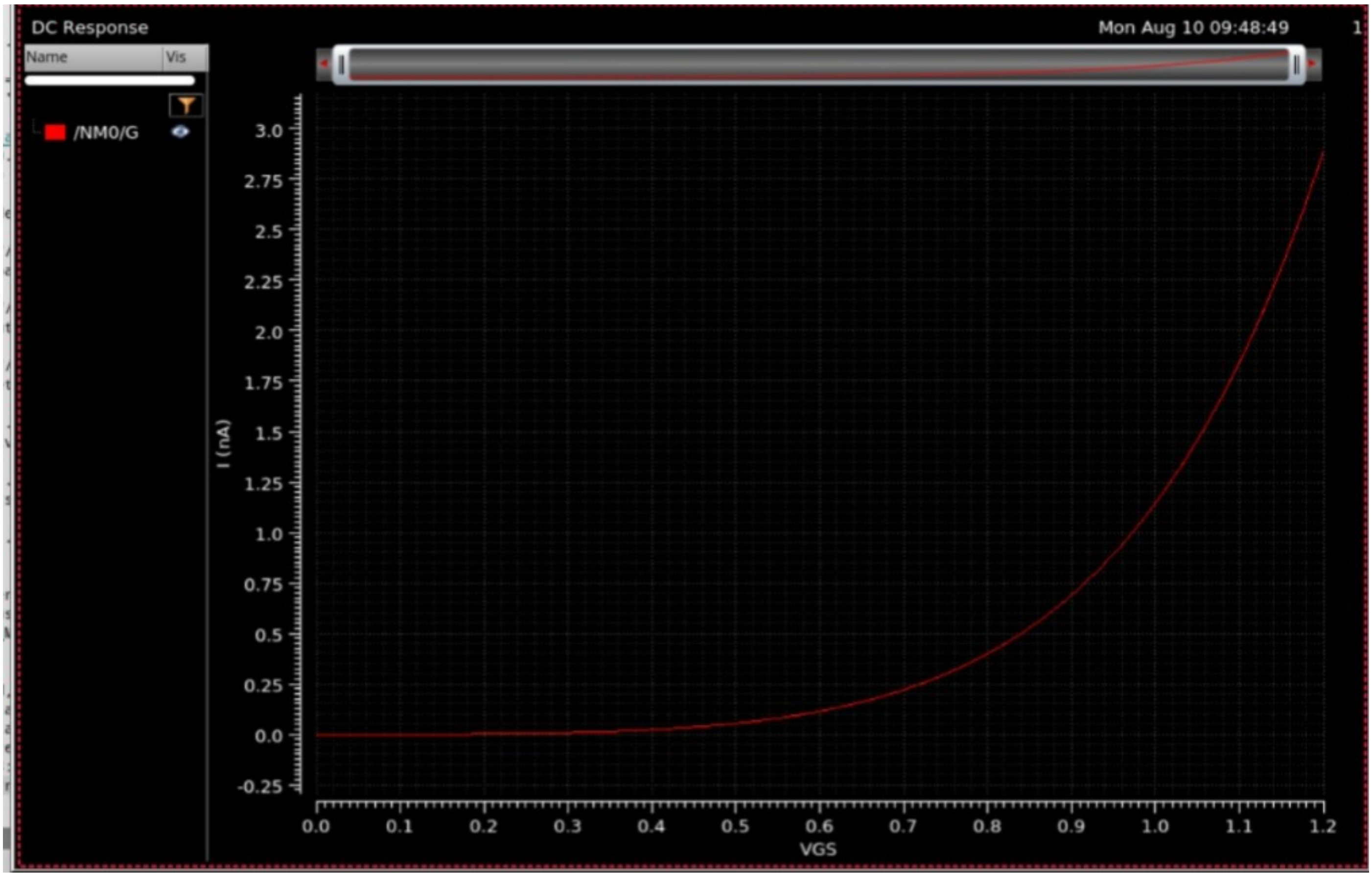
Device dimensions: W = 1 µm, L = 100 nm. VGS and VDS are applied through the voltage sources.

PARAMETRIC SWEEP

VGS sweep range: 0 V to 1.2 V

VGS Transfer Characteristic

Drain current response versus gate-source voltage



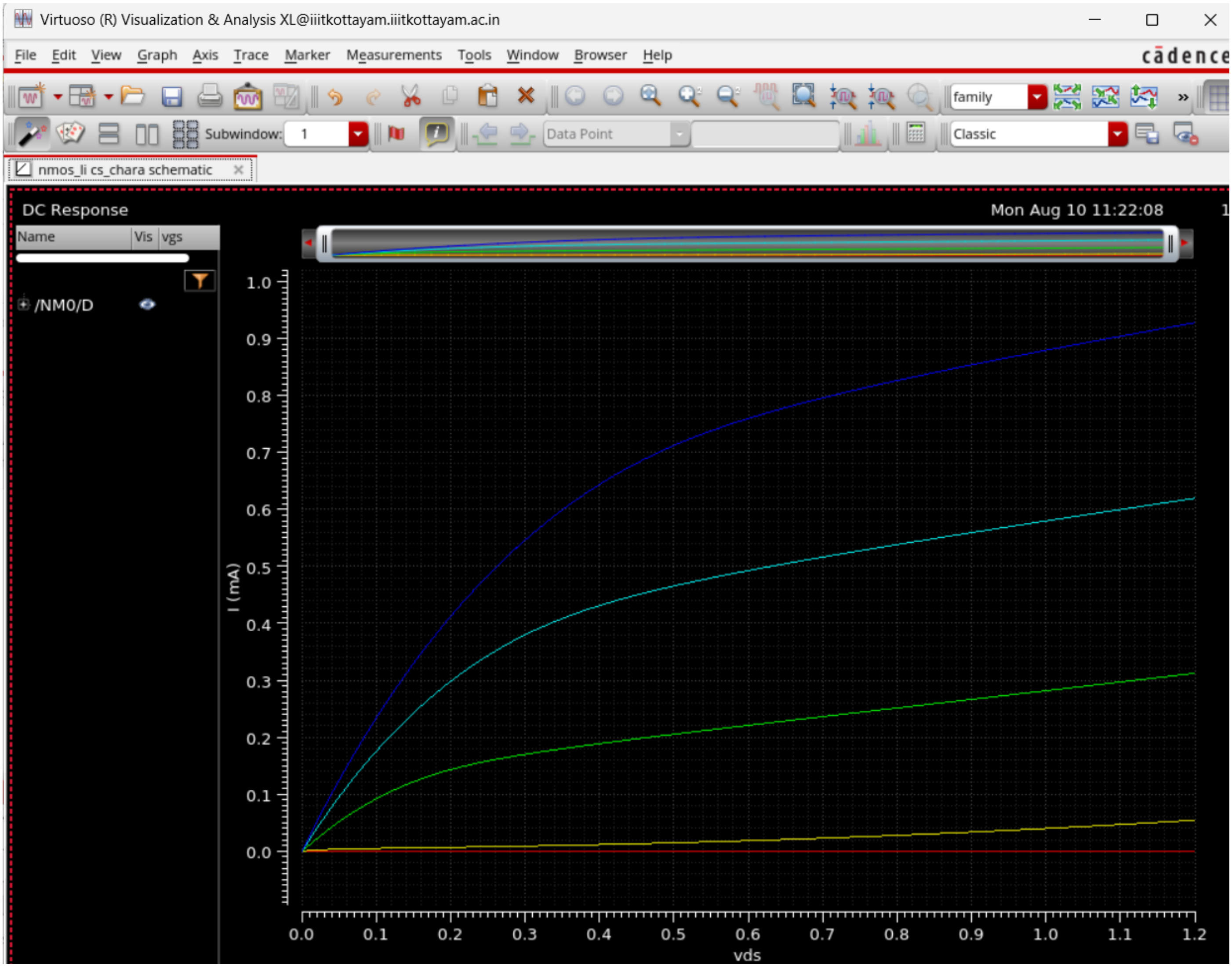
DC response showing drain current increasing as VGS is increased.

OBSERVATION

The current remains very small at low gate-source voltage and rises strongly as VGS increases.

VDS Output Characteristics

Parametric sweep for multiple VGS values



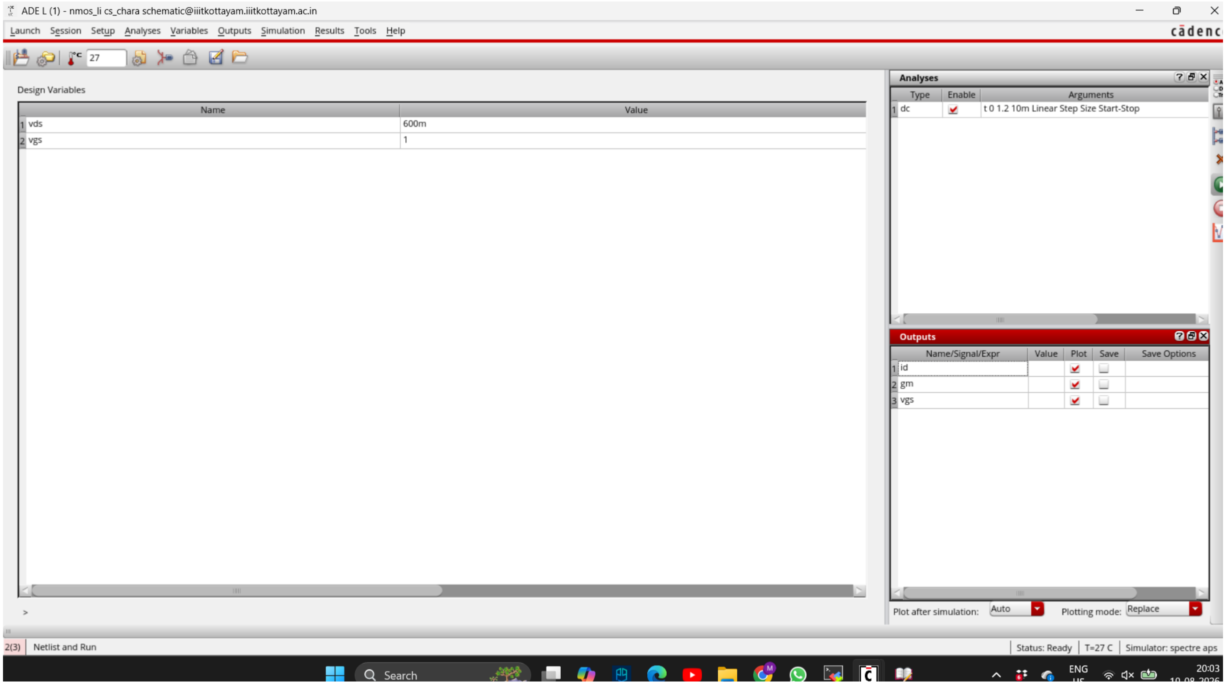
Drain current ID plotted against VDS for multiple VGS values.

OBSERVATION

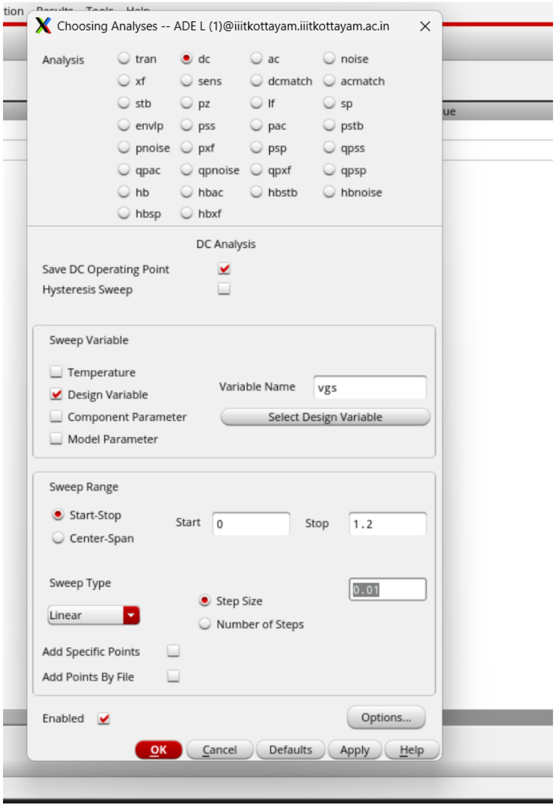
Higher VGS produces higher drain current, while the curves become less steep as VDS increases.

Simulation Setup

DC analysis configuration and design variables



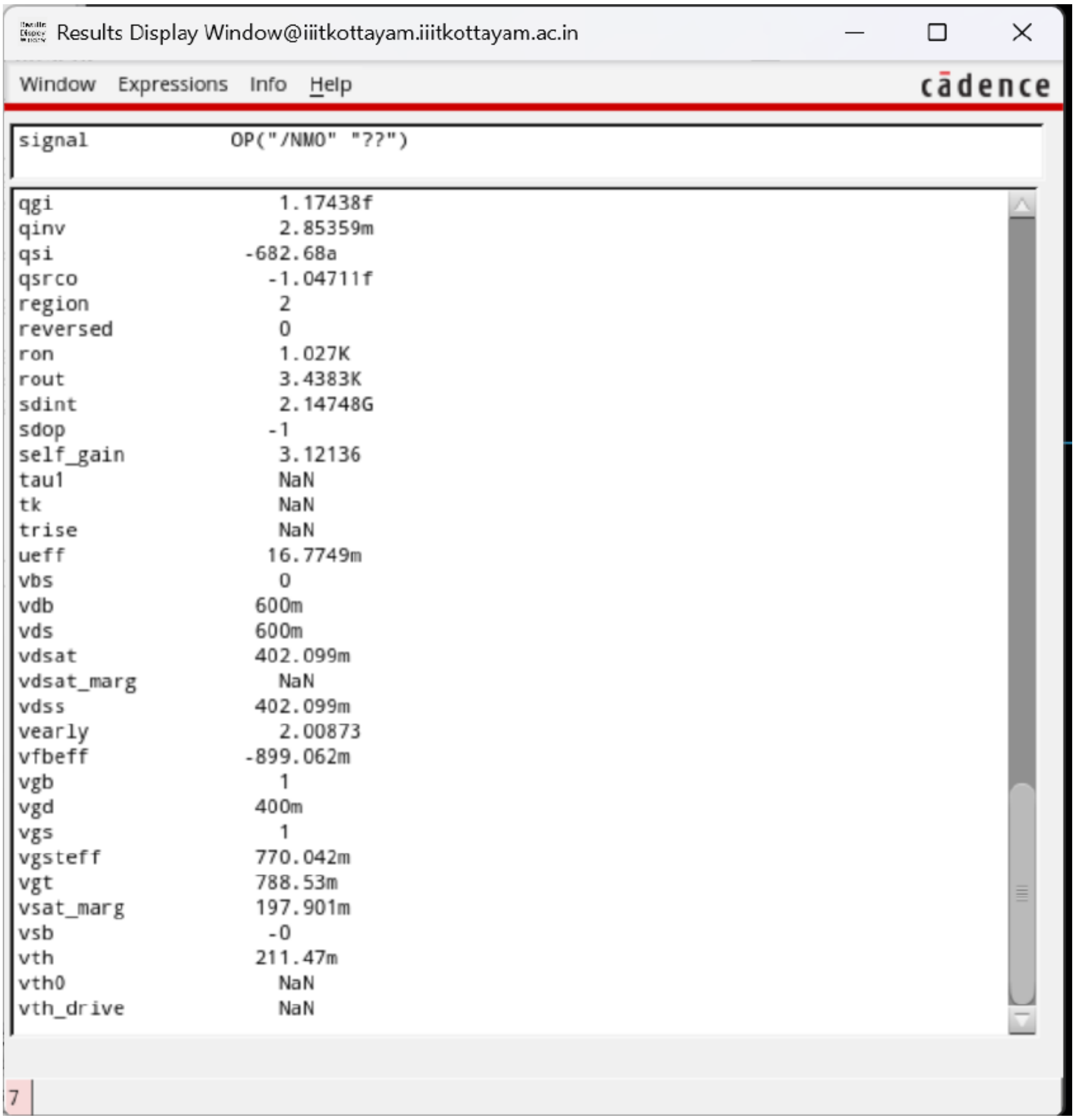
ADE L setup showing VDS and VGS design variables and the enabled DC analysis.



DC analysis configuration with VGS swept from 0 V to 1.2 V.

DC Operating Point

Spectre operating-point results



Spectre operating-point result window for the NMOS device.

KEY RESULT

VDS = 0.6 V • VGS = 1 V

Reported threshold voltage: approximately 211.47 mV.